

Your Digital Employee

DigitalRM is a personal digital employee — named **Aria** — that does your work for you. You delegate in plain language; she captures tasks, keeps your notes and contacts, drafts and researches, reminds you of what's due, escalates the calls that are yours to make, and reports back. Built on a typed multi-database foundation with tenant isolation, at-rest encryption, and a full, model-attributed audit trail.

● **TASKS** ● **NOTES & KNOWLEDGE** ● **CONTACTS** ● **PROACTIVE REMINDERS**
● **GROUNDING ANSWERS** ● **AUDITED** ● **PLUGGABLE AI**

SCOPE

One trusted assistant per person — owns your plate, your knowledge, and your follow-ups end to end

CORE

Reason, act through tools, ground on your notes, remind, escalate, and record every decision

DELIVERY

Web chat, HTTP API, and CLI today; Telegram / WhatsApp / Slack plug into the same engine

EXTENSIBILITY

New skills, channels, and entities add cleanly; the reasoning brain is swappable per env

HOW ARIA WORKS — THE AGENTIC LOOP

Understand

Read the request and the live thread; resolve who and what it's about from your notes and contacts.

Plan

Decide the next step on the reasoning tier — which tool, with what arguments, grounded in what's known.

Act

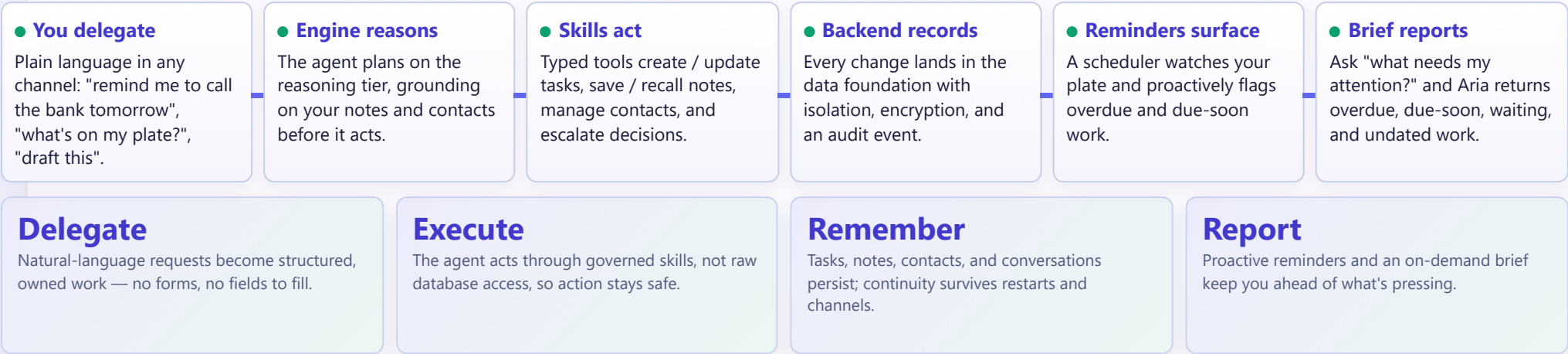
Call a typed skill: create or update a task, save or recall a note, look up a person, or escalate to you.

Reflect

Read the result, loop until the work is done, then report back — and record the decision and its model.

A colleague that takes work off your plate — not another app to manage.

A digital employee sits between you and your systems. You hand it work the way you would a capable assistant; it understands intent, acts through governed tools, keeps the details so you don't have to, and closes the loop. DigitalRM is that layer, built to be trusted with real action — every write is tenant-scoped, permission-checked, and auditable.



Anything you would hand to a reliable assistant fits: capture and chase tasks, keep a second brain of notes and decisions, track the people you deal with, draft replies, set reminders, and surface what needs you — with a clear record of what was done and why.

THE JOB	WHAT DIGITALRM PROVIDES	OUTCOME FOR YOU
Run your plate	Task capture, status, priority, due dates, and a live "active work" view through governed skills.	Nothing falls through the cracks; the backlog is one query, not a memory exercise.
Be your memory	Durable notes and contacts with semantic recall, distilled from conversation into a searchable knowledge base.	Answers are grounded in what you actually told it, not guessed.
Stay ahead	A scheduler that surfaces overdue and due-soon work, plus an on-demand "what needs my attention" brief.	You're reminded before things slip, and can open the day with one question.

What Aria does — the work she owns on your behalf.

Each capability is a typed skill the agent can call. The model never touches the database directly; it only invokes these governed verbs, and every consequential one is audited.

TASKS & YOUR PLATE

Capture, update, prioritise, and date work across nine kinds — draft, research, follow-up, reminder, scheduling, errand, admin, finance, review. List the live plate on demand.

NOTES & KNOWLEDGE

Save durable facts, decisions, and documents into collections; recall them semantically. The canonical text lives in Postgres, the meaning-vector in Qdrant.

CONTACTS

Track the people you deal with — colleagues, clients, vendors, family — with relationship and profile. Contact PII is encrypted at rest and masked in logs.

REMINDERS & BRIEF

A background scheduler flags overdue and due-soon work; the brief returns overdue, due-soon, waiting-on-you, and undated work in one structured view.

DRAFTING & RESEARCH

The reasoning tier composes replies, summaries, and short research, attached to the relevant task — prepared for your review, not sent on your behalf.

ESCALATION TO YOU

Money, outward messages, and irreversible or ambiguous calls are handed back to you with a clear reason, never improvised. Reversible internal work she does freely.

MEMORY & CONTINUITY

Conversations and messages persist per channel and thread, so any surface resumes the same context and continuity survives restarts.

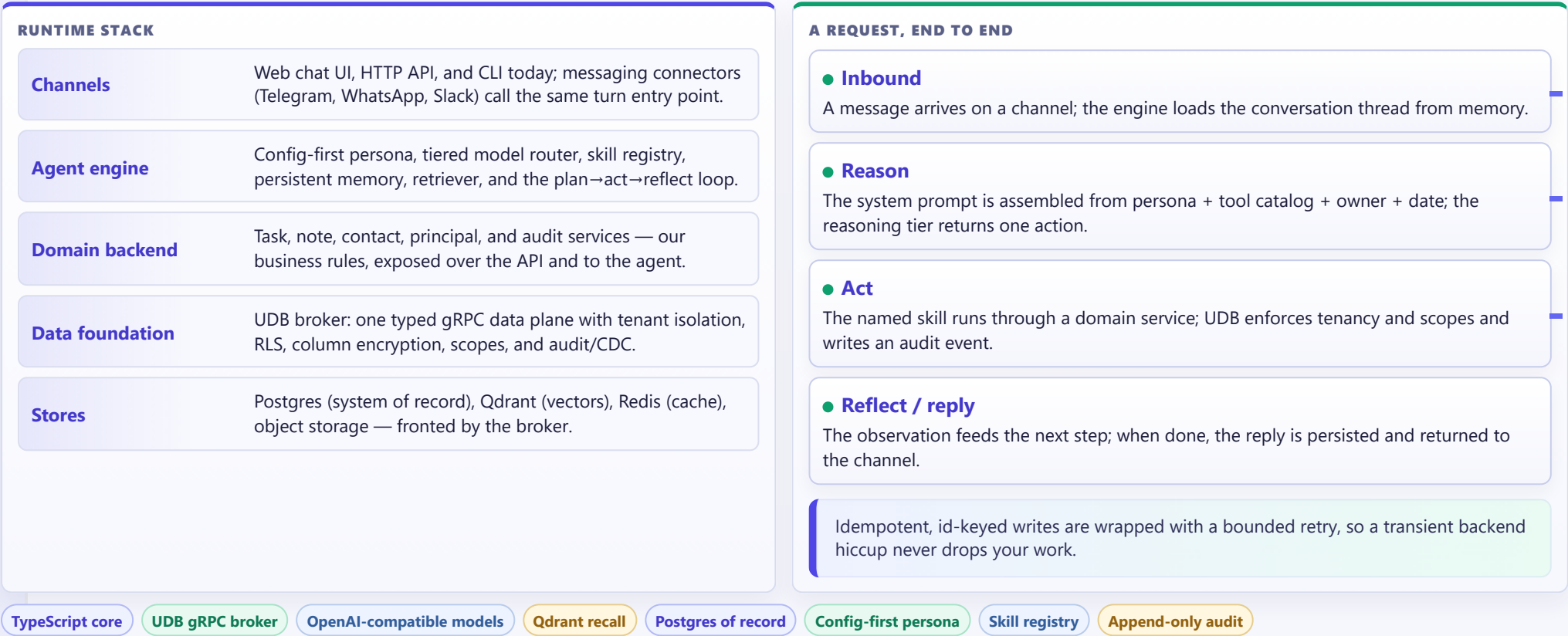
AUDIT & EXPLAINABILITY

Every consequential action writes an append-only audit event with the action, outcome, rationale, and the exact model that drove the decision.

SKILL FAMILY	REPRESENTATIVE VERBS	GOVERNED BY
Work	create_task, update_task, list_tasks, brief, escalate	Tenant scope + write scope; status & kind validation; audit on every write.
Knowledge	save_note, search_notes (semantic)	Tenant scope; embeddings indexed on save; recall filtered to your workspace.
People	lookup_contact, upsert_contact	PII scope to read encrypted fields; column-level encryption at rest.

Four clean layers: channels, our agent engine, our backend, and a data foundation.

DigitalRM is our own code on a proven foundation. The data/identity/storage layer is the UDB broker; the backend services and the agent engine are ours. Nothing above the data layer touches the database SDK directly — the agent calls skills, the skills call domain services, the services call UDB.



Our own reasoning runtime — character, skills, memory, and a deliberate loop.

The engine stitches the ideas worth keeping from leading agent runtimes into our code: a character defined in editable Markdown, a registry of typed skills, durable memory, a model router, and an explicit agentic loop. No black-box wrapper — every part is inspectable.

PERSONA (CONFIG-FIRST)

Who Aria is and how she works lives in SOUL / AGENTS / SKILL / MEMORY Markdown, loaded at startup. Editing the files changes behaviour with no code change or redeploy.

MODEL ROUTER

Two tiers: a primary tier for planning and judgement, a fast tier for routing and short replies. Both are any OpenAI-compatible endpoint, chosen by environment.

SKILL REGISTRY

Tools are typed calls into domain services. The model sees only the skill catalog and can never reach the database — a hard boundary between reasoning and action.

SESSION MEMORY

Conversations and messages are stored in the backend, so every channel resumes the same thread and continuity survives restarts.

RETRIEVER

Grounding over your notes through a clean interface — semantic recall via embeddings + Qdrant, with a keyword fallback when no embedding provider is set.

THE LOOP

Plan → act → observe → reflect, bounded by a step budget. Each step emits a thought, a tool call, and an observation, ending in a final reply.

One turn, in the agent's own protocol

STEP	WHAT HAPPENS	DETERMINISM
Reason	The model replies with exactly one JSON object: either a tool call {thought, action, args} or a finish {thought, final}.	Strict protocol; malformed output is treated as a final reply rather than crashing the turn.
Act	The named skill executes; its return value becomes the observation the model sees next.	Unknown tools and tool errors return a structured error the model can recover from.
Loop	Observations are appended and the model decides the next step, up to the step budget.	Bounded — the turn always terminates with a reply, never an open-ended run.

A pluggable brain — grounded, tiered, and provider-neutral.

The reasoning model is not hard-wired. Any OpenAI-compatible endpoint works, selected per environment, so you choose the trade-off between depth, latency, cost, and where inference runs. Answers are grounded on your own notes before they are asserted.

DEFAULT BRAIN
NVIDIA Nemotron via NIM when a key is present — open, agentic-reasoning models served OpenAI-compatible. The development default is qwen3.6-plus.

TIERED ROUTING
Primary tier for planning, judgement, and customer-facing wording; fast tier for classification, routing, and short confirmations.

GROUNDING ANSWERS
Factual replies are backed by semantic recall over your notes (embeddings → Qdrant), so Aria quotes what you told her instead of inventing.

MODEL ROLE	USED FOR	HOW IT'S SET
Primary (reasoning)	Planning the next action, judgement, and the final message to you.	Provider + model via env; defaults to Nemotron (NIM) or qwen3.6-plus.
Fast	Cheap routing, classification, and short acknowledgements.	Independent env override; falls back to a smaller model on the same provider.
Embeddings	Vectorising notes on save and queries at recall time.	OpenAI-compatible embeddings endpoint; default text-embedding-v4.

Swapping the brain is one environment variable. The same agent loop, skills, persona, and grounding run unchanged whether the model is hosted by NVIDIA, a cloud provider, or on your own infrastructure.

An employee, not a database — she watches your plate and surfaces what's pressing.

A passive task list waits for you to check it. DigitalRM does the opposite: a scheduler runs on an interval, finds work that is overdue or coming due, and pushes a reminder — and you can ask for a brief at any time to open your day.

PROACTIVE REMINDERS

- A background scheduler checks the live plate on a configurable interval.
- Overdue and due-soon tasks are surfaced through a notify hook — stdout today, a real channel (Telegram / WhatsApp) by plugging into the same hook.
- Each reminder is de-duplicated, so you are never pinged twice about the same task.
- Due dates are stored as sortable ISO text and resolved from natural language ("tomorrow", "in 3 days") on the way in.

THE BRIEF — "WHAT NEEDS MY ATTENTION?"

- **Overdue** — past their due date and still open.
- **Due soon** — within the next window (default 24 hours).
- **Waiting on you** — blocked items escalated for your decision.
- **Undated** — active work with no date yet.

The web chat greets you with the brief on load; the agent answers "what needs my attention?" by calling the same brief skill.

Watch

Interval scheduler over the active plate, started with the server.

Surface

Overdue and due-soon reminders via a pluggable notify hook.

Brief

On-demand, structured view of everything pressing.

Act

From a reminder you can reschedule, close, or escalate in the next turn.

Built to be trusted with real action — isolation, encryption, scopes, and a record.

A digital employee only earns delegation if it is safe by construction. DigitalRM enforces its guarantees server-side in the data foundation, not as application convention, and keeps a human in the loop for the decisions that are yours.

TENANT ISOLATION

Every read and write is scoped to your workspace server-side. A request can never reach another tenant's data — cross-tenant access returns "not found" by design.

PII ENCRYPTION

Contact and owner email/phone are encrypted at rest (AEAD) and masked in logs. Reading them back requires a dedicated PII scope — held only where it's needed.

LEAST PRIVILEGE

The agent carries only the scopes it needs. A denied permission stops the action cleanly rather than being retried — capability is explicit, not assumed.

HUMAN IN THE LOOP

Money, outward messages, and irreversible or ambiguous actions are escalated back to you with a reason. Aria prepares them fully but does not act in your name.

CONTROL	WHERE IT'S ENFORCED	WHAT IT PREVENTS
Tenant & project isolation + RLS	UDB broker, on every Select / Upsert / Delete.	Any cross-workspace data leak, on reads and writes alike.
Column encryption + PII scope	Storage layer + scope check at the data plane.	Plaintext PII at rest, and unscoped reads of sensitive fields.
Append-only audit, model-attributed	Domain services, on every consequential action.	Unexplained actions — every change records who, what, why, and which model.
Escalation policy	Agent operating rules + an escalate skill.	The agent committing money or outward actions without your approval.

One typed data plane in front of every store — the foundation we build on.

Rather than wire the agent to many database clients, DigitalIRM sits on the UDB broker: a proto-driven, multi-database gRPC data plane. The domain is declared as annotated schema and the broker handles routing, isolation, encryption, migrations, and an event stream.

WORKSPACE ENTITIES (DIGITALRM.V1)

ENTITY	HOLDS
Principal	The owner Aria works for — identity, timezone, preferences.
Task	Your plate — kind, status, priority, due date, structured payload.
Note	Your knowledge — collection, title, body, and a recall vector.
Contact	Your people — relationship, org, encrypted email/phone.
Conversation / Message	Omnichannel session memory, per channel and thread.
AuditEvent	Append-only record of every consequential action.

WHAT THE FOUNDATION GIVES US

Typed API

One gRPC data plane addressed by message type — Select / Upsert / Delete — instead of N database drivers.

Isolation

Tenant and project scoping with row-level security, enforced fail-closed on every call.

Encryption

Column-level at-rest encryption for PII, with scope-gated decryption.

Events

Change-data-capture topics per entity for downstream sync, analytics, and integrations.

Native services

Auth, storage, and notifications available from the same control plane.

The agent never holds raw database credentials — only a tenant-scoped token. The store credentials live with the broker, so the blast radius of the agent layer is small by design.

One engine, many surfaces — and a clean API to build on.

Every channel is a thin adapter over a single entry point: give the engine a message, a channel, and a thread, and it runs a turn. Web chat, CLI, and an HTTP API ship today; messaging connectors reuse the same path and the same memory.

WEB CHAT
A built-in browser UI that greets you with your brief on load and streams the agent's tool use as it works.

HTTP API
A small JSON API for the agent and the domain — embed Aria in your own app, intranet, or workflow.

CLI
Drive a turn from the terminal or a script; ideal for automation and quick checks.

ENDPOINT	PURPOSE
POST /api/chat	Run one agent turn — message in, grounded reply and tool-trace out.
GET /api/brief	Structured "what needs my attention": overdue, due-soon, waiting, undated.
GET /api/reminders	Recent proactive reminders surfaced by the scheduler.
GET / POST /api/tasks · /api/notes · /api/contacts	Direct domain access for dashboards and integrations.

Adding a messaging channel (Telegram, WhatsApp, Slack) is an adapter that forwards inbound messages to the engine and delivers the reply — persona, memory, skills, and grounding are shared, so behaviour is identical everywhere.

Portable to run, simple to operate, configured by environment.

The core is a single Node service; the foundation is the broker plus its stores. Everything is set by environment, so the same build runs on a laptop, a server, or in containers — with managed or self-hosted backends.

CORE SERVICE
One Node process serving the web chat, the API, the agent engine, and the reminder scheduler. Starts with a single command.

FOUNDATION
The UDB broker fronts Postgres, Qdrant, Redis, and object storage — local containers for development, managed services in production.

CONFIGURATION
Connection strings, model keys, scopes, and the reasoning brain are all environment variables; no secrets in code or images.

REPRODUCIBLE BRING-UP
Scripted setup brings up services, applies the schema, provisions the workspace identity, and starts the broker end to end.

Operational properties

RESILIENT WRITES
Idempotent, id-keyed upserts with bounded retry absorb transient backend errors without duplicating work.

VERIFIED BEHAVIOUR
A scripted "day with Aria" exercises every capability and asserts the resulting backend state, not just the wording.

OBSERVABLE
Health and brief endpoints, model-attributed audit, and reminder logs make the running system easy to inspect.

The full picture — what DigitalRM brings across the workday.

DigitalRM combines a trustworthy data foundation, a transparent agent engine, and a proactive operating model into one personal employee you can actually delegate to.

CATEGORY	CAPABILITIES	BACKED BY
Work management	Task capture, status, priority, due dates, nine work kinds, live plate, escalation to the owner.	Task service, governed skills, audit.
Knowledge	Durable notes, collections, semantic recall, grounded answers, second-brain continuity.	Note service, embeddings, Qdrant.
People	Contacts with relationship and profile, encrypted PII, scoped read access.	Contact service, column encryption.
Proactivity	Interval reminders, overdue and due-soon surfacing, on-demand brief, day-opening greeting.	Scheduler, brief skill, notify hook.
Reasoning	Tiered model router, plan→act→reflect loop, strict tool protocol, pluggable provider.	Agent engine, OpenAI-compatible models.
Trust & governance	Tenant isolation, RLS, at-rest encryption, least-privilege scopes, model-attributed audit, human-in-the-loop.	UDB broker, domain services, operating rules.
Channels & API	Web chat, CLI, HTTP API; pluggable messaging connectors over one engine entry point.	HTTP server, channel adapters.
Operations	Single Node service, broker + stores, env-driven config, scripted bring-up, resilient writes.	Core runtime, scripts, Docker.

PRODUCT STATEMENT

DigitalRM is a personal digital employee — a trustworthy, proactive assistant that owns your tasks, notes, contacts, and follow-ups, and acts through governed, audited tools.

ENGINEERING STATEMENT

A typed UDB data foundation, our own domain backend, and our own agent engine — config-first persona, tiered models, skills, memory, semantic recall, and a deliberate plan→act→reflect loop.